Arista Access Switch Hardware/Environmental Fault SOP for WHI-ACC-61 at Whitefield East POP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6671

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults on the Arista Access Switch, specifically the ASIC memory parity errors, line card crashes, and power supply redundancy loss on Node ID WHI-ACC-61 located at Whitefield East POP. The failure domain occurs due to the intricate relationship between the ASIC memory parity checks, line card power management, and power supply redundancy mechanisms. In the Arista Access Switch, the ASIC memory parity checks are performed by the ASIC's built-in memory controller, which generates a parity bit for each memory byte to detect single-bit errors. However, in the presence of multi-bit errors or power supply fluctuations, the ASIC memory parity checks may fail, leading to line card crashes or power supply redundancy loss. Furthermore, the power supply redundancy mechanism is designed to provide uninterrupted power supply to the line cards. However, in cases where the power supply redundancy is lost, the line cards may crash or experience ASIC memory parity errors, resulting in the failure domain.

Pre-requisite Checks

- Verify the dispatch of remote hands to the Whitefield East POP for physical access to the affected equipment.

- Check the redundant link status to ensure that it is operational and not contributing to the failure domain.

- Verify the power supply redundancy status to ensure that it is functioning as expected.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal hardware cpu-memory, Expected Output: CPU memory information, including memory parity errors and misfires., Description: This command displays the CPU memory information, including memory parity errors and misfires, which can help identify the root cause of the failure domain.
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```
Command: show system internal hardware linecards, Expected Output: Line card information, including status and errors., Description: This command displays the line card information, including status and errors, which can help identify the root cause of the failure domain.

Command: show system internal hardware power-supply, Expected Output: Power supply information, including status and redundancy., Description: This command displays the power supply information, including status and redundancy, which can help identify the root cause of the failure domain.
```

Resolution Steps

Step 1: Step: Reset the ASIC Memory Parity Checks, Command: clear system internal hardware cpu-memory, Description: This command resets the ASIC memory parity checks, which can help resolve the memory parity errors and misfires.

Step 2: Step: Replace the Faulty Line Card, Command: reload linecard , Description: This command reloads the line card, which can help resolve the line card crashes and ASIC memory parity errors.

Step 3: Step: Restore Power Supply Redundancy, Command: enable system internal hardware power-supply redundancy, Description: This command restores the power supply redundancy, which can help resolve the power supply redundancy loss and associated failures.

Post-Check Verification

```
Command: show system internal hardware cpu-memory, Expected Output: CPU memory information, including memory parity errors and misfires., Description: This command verifies that the ASIC memory parity checks are functioning as expected.

Command: show system internal hardware linecards, Expected Output: Line card information, including status and errors., Description: This command verifies that the line card crashes and ASIC memory parity errors have been resolved.

Command: show system internal hardware power-supply, Expected Output: Power supply information, including status and redundancy., Description: This command verifies that the power supply redundancy has been restored.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: rollback , Description: This command restores the original configuration, which can help roll back any changes made during the resolution process.

- Step: Revert to the Previous Software Image, Command: boot system internal software previous-version, Description: This command reverts to the previous software image, which can help roll back any changes made during the resolution process.